PVC Suspension Reactor Ver 1 · R-101 (one of four 140 m³ batch suspension-PVC reactors)

Relief load basis

PVC-V1-PE-RLB-001

Relief load basis of R-101 generated from the OTS model. Governing case: Jacket temperature control fails to full steam (TV-110A wide open) at the design state (SIS assumed to fail), 346 t/h of VCM vapour at the 15.0 bar g set → PSV-100 2 × R. Runaway relief is two-phase; the vapour areas here are a lower bound until the DIERS study. Checks: 17 ✓ / 4 ✗; 8 engineering queries.

Unit	PVC Suspension Reactor Ver 1 · R-101 (one of four 140 m³ batch suspension-PVC reactors)
Prepared for	Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University
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Originator	Apinan Soottitantawat

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The numbers in this document come from the PVC Suspension Reactor Ver 1 simulator model (R-101) and the documents of the same OTS, or are labelled "assumed" or "typical". It was generated for teaching: it is not a design deliverable and must not be used for construction or purchase. Checked and Approved are left blank on purpose.

1 Purpose and basis

This relief load basis lists the overpressure scenarios of API 521 for reactor R-101, identifies the governing case for each relief device, sizes PSV-100 and TSV-110 with API 520, checks the inlet and outlet piping, and describes the emergency vent. It is generated from the R-101 simulator model: model relief runs on copies of the model with the SIS switched off, the model constants and the other OTS documents. Instrumented protection is not credited (typical practice).

Protected equipment	R-101, 141.5 m³, MAWP 15.2 bar g / 93 °C; jacket loop 5.2 bar g (P&ID-001 note 3)
Relief device	PSV-100 set 15.0 bar g to D-103 (PFD-100, P&ID-001)
Governing case	Jacket temperature control fails to full steam (TV-110A wide open) at the design state — 346 t/h vapour at the set
Selected orifice	2 × R (6 × 8 in), 19182 mm² required, capacity 372 t/h
Limits (typical)	set ≤ MAWP; accumulation ≤ 10 % (single valve), 16 % (multiple valves), 21 % (fire); inlet loss ≤ 3 %; built-up back-pressure ≤ 10 % (conventional) / 50 % (balanced bellows)

Runaway relief of a VCM suspension is two-phase (DIERS). The OTS model relieves vapour only (step(): nPsv = 0.25·√(P − 1.2) kmol/s, P in bar a), so every area in this document is a lower bound until the two-phase study (HAZOP R08) is done.

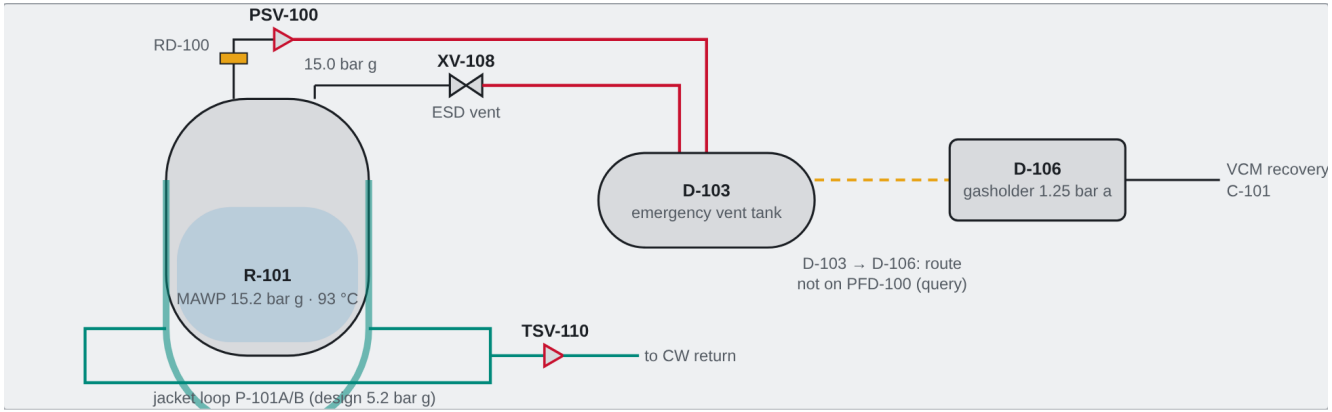


Figure 1 · Relief and vent system of R-101 (sketch; RD-100 and TSV-110 are recommended items of the Safety layers register).

2 Overpressure scenarios

Scenario (API 521)	Mechanism on R-101	Quantified by	Phase	Result	Load, t/h	Status
Runaway: total loss of cooling water (SIS assumed to fail)	CW header lost: jacket injection, baffles and E-101 stop removing heat; Q_{rx} rises ≈ 9.4 %/K	Model run 3.1 (design state)	Vapour in the model; two-phase in practice (DIERS)	lifts 35:34 after the failure; peak 15.72 bar g; model relief ≤ 221.7 t/h	270.9	covered
Runaway: total loss of cooling water at the start of the reaction	As above, earlier in the batch (more monomer)	Model run 3.2	Vapour (model) / two-phase	lifts 59:03 after the failure; peak 15.00 bar g; model relief ≤ 216.5 t/h	201.7	covered
Abnormal heat input: jacket temperature control fails to full steam	TIC-110 output to 0 %: TV-110A LP steam 100 %, TV-110B CW 0 %; the jacket heats the batch	Model run 3.3	Vapour (model) / two-phase	lifts 17:12 after the failure; peak 18.71 bar g; model relief ≤ 242.1 t/h	345.9	governs

Scenario (API 521)	Mechanism on R-101	Quantified by	Phase	Result	Load, t/h	Status
Utility failure: electric power	Agitator M-101, jacket pumps P-101A/B and P-103 stop; CW and air stay	Model run 3.4	Vapour (model) / two-phase	lifts 77:32 after the failure; peak 15.00 bar g; model relief ≤ 216.5 t/h	252.4	covered
Loss of agitation	M-101 stops; jacket circulation and CW stay	Model runs 3.5 and 3.6	—	does not reach the set (peak 8.68 bar g, 57.3 °C in 180 min); near the gel peak (4.0 h) it does not reach the set (peak 8.82 bar g, 58.8 °C in 120 min). The lumped, well-mixed model cannot show a separated monomer layer (SRS SIF-2).	0	not reached (model limit)
External fire (pool fire under R-101)	Heat input Q = 43.2·F·$A_{ws}^{0.82}$ kW on the wetted area up to 7.6 m above grade	Calculator 4.3	Vapour	A_{ws} 56.9 m² (bottom 4.0 m above grade, assumed), Q 1188 kW, λ 229 kJ/kg at 18.39 bar g	18.7	covered
Blocked outlet (vapour)	R-101 has no continuous vapour outlet while reacting (vents and degassing are closed); a closed vent adds no load beyond the runaway	Argument	—	No separate load	0	not applicable
Control-valve failure on the VCM charge (FV-102 wide open)	Source is P-103: shut-off 12.0 bar g (model K.Pvcm 13.0 bar a) is below the 15.0 bar g set	Model constants	Liquid	No vapour overpressure; 225 t/h at the design pressure would overfill — stopped by the FQI-102 preset and LSHH-100 (overfill: next row)	0	not credible (source below set)
Liquid overfill: shrinkage-water valve FV-103 wide open	≈ 10.0 m³/h of water until the vessel is liquid-full; then the injection pump shut-off limits the pressure	Model run 3.7	Liquid	liquid-full (level 96.1 %), peak 14.98 bar g = injection pump shut-off 14.99 bar g (K.Pinj), just below the set: no lift	0 (liquid case in 4.2)	covered by the pump shut-off (margin small)
Nitrogen regulator failure (PCV-141 pad)	N_2 header assumed 8 bar g, below the set	Argument	Gas	Pressure rises to the N_2 header at most; PIAH-100/PSHH-101 see it. The N_2-pressurised short-stop pot D-105 must sit above the trip pressure to inject: its pressure and restriction orifice are not stated (query)	0	not credible for R-101; D-105 open
Thermal expansion of the blocked-in jacket	Jacket isolated full of water, heated by the batch (no circulation)	Calculation 4.4	Liquid	151 L/h; pressure rises 0.092 bar/s → 5.2 bar g in 40 s	0.15 (jacket)	TSV-110
Instrument-air failure	Control valves go to their fail positions: TV-110A closes, TV-110B, FV-120, FV-130 open (more cooling)	P&ID fail positions	—	No load; XV-104 and XV-108 cannot open without air (HAZOP R39)	0	not applicable
Back-flow from VCM recovery / gasholder	Recovery at 1.25 bar a	Model constant	Gas	Source below the set	0	not credible

2.1 Worked calculation of the governing load and area

Governing run: **Jacket temperature control fails to full steam (TV-110A wide open) at the design state**, SIS off. When PSV-100 lifts (17:12 after the failure) the model has X = 0.378, T = 80.3 °C, heat release Q_{rx} = 20986 kW and heat removal (jacket + baffles + E-101) = −2047 kW (negative: the jacket is heating). For a tempered system the excess heat must leave as latent heat of the boiling monomer:
$W = (Q_{rx} + 0.95\,Q_{agitator} - Q_{removed} - Q_{ambient}) / \lambda(T) \approx 23033$ kW / 242.4 kJ/kg × 3600 = **345.9 t/h**
(agitator and ambient terms from step() are included in the run value). Relieving conditions: set 15.0 bar g + 10 % = 16.50 bar g = 1751 kPa a; T_1 = 85.4 °C where VCM + water vapour pressure equal P_1 (DIPPR and Antoine curves of the model); M 62.5, k 1.18 and Z 0.82 (assumed), K_d 0.975, K_b 1, K_c 0.9 (rupture disc RD-100):
C = 0.02545, A = 345900 / (0.02545 × 0.975 × 1751 × 1 × 0.9) × √(358.5 × 0.82 / 62.5) = **19182 mm²** → API 526 **2 × R** (6 × 8 in, 20646 mm²), capacity 372 t/h.
For comparison the model's PSV curve passes 227 t/h at the same pressure (≈ 12599 mm²).

3 Relief register

Tag	Protects	Set, bar g	Governing scenario	Load, kg/h	Phase	Area, mm²	Orifice	Discharge to	Notes
PSV-100	R-101 reactor (MAWP 15.2 bar g / 93 °C)	15.0	Jacket temperature control fails to full steam (TV-110A wide open) at the design state (governing; SIS assumed to fail)	345900	Vapour (model, tempered VCM); two-phase per DIERS to be confirmed	19182	2 × R (6 × 8 in)	D-103 emergency vent tank (closed system)	Relieving 16.50 bar g, 85.4 °C; conventional, rupture disc RD-100 upstream (Kc 0.9); model relief up to 242 t/h; Runaway two-phase sizing TBC (DIERS, R08)
RD-100	PSV-100 inlet (polymer plugging)	15.0	As PSV-100 (in series)	345900	As PSV-100	—	Disc ≥ PSV inlet size	PSV-100	Recommended (HAZOP R35): burst ≤ PSV set, pressure-monitored interspace; combination factor Kc 0.9 (typical, API 520)
TSV-110	Jacket loop (design 5.2 bar g)	5.2	Thermal expansion of the blocked-in jacket	150	Liquid (water)	2.2	D (1 × 2 in)	CW return header (closed)	Recommended (Safety layers); the blocked-outlet case needs more (query: 396 t/h CW injection → Q)

4 Model relief runs

Run settings. Copies of the R-101 model, SIS off (pvBatch noTrip), dt 0.25 s, noise off, all other controllers in AUTO, PSV-100 as in step(): nPsv = 0.25·√(P − 1.2) kmol/s (P bar a) while open, opens above 15.0 bar g, recloses below 14.0 bar g. Times are from the failure.

Run	Start	Failure	TSHH point	PSHH point	PSV lift	Open, s	Peak P, bar g	Peak T, °C	Max relief, t/h	Relieved, t	Load at set, t/h
3.1 CW loss · design	design state (2.0 h into the reaction)	CW header flow −95 % (jacket injection, baffles, E-101) at t = 0	20:35	31:34	35:34	243	15.72	106.3	222	14.7	271
3.2 CW loss · start	start of the reaction	CW header flow −95 % at the initiator charge	37:56	53:51	59:03	444	15.00	102.8	217	26.3	202
3.3 TV-110A full open	design state (2.0 h into the reaction)	TIC-110 output 0 % → split range sends TV-110A (LP steam) 100 %, TV-110B (CW) 0 %	8:33	14:22	17:12	305	18.71	112.6	242	19.2	346
3.4 Power failure	design state (2.0 h into the reaction)	M-101, P-101A/B and P-103 stop at t = 0 and stay off; CW and air remain	55:02	72:48	77:32	72	15.00	97.7	217	4.3	252
3.5 Agitator stops · design	design state (2.0 h into the reaction)	M-101 stops at t = 0; jacket loop and CW remain	—	—	not reached	—	8.68	57.3	—	—	—
3.6 Agitator stops · 4.0 h	reference batch, failure at 4.0 h	reference batch from the start; M-101 stopped at 4.0 h	—	—	not reached	—	8.82	58.8	—	—	—
3.7 FV-103 wide open	start of the reaction	FIC-103 in MAN at 100 % for the whole batch (≈ 10 m³/h of water at 30 °C)	—	176:10	not reached	—	14.98	57.4	—	—	—

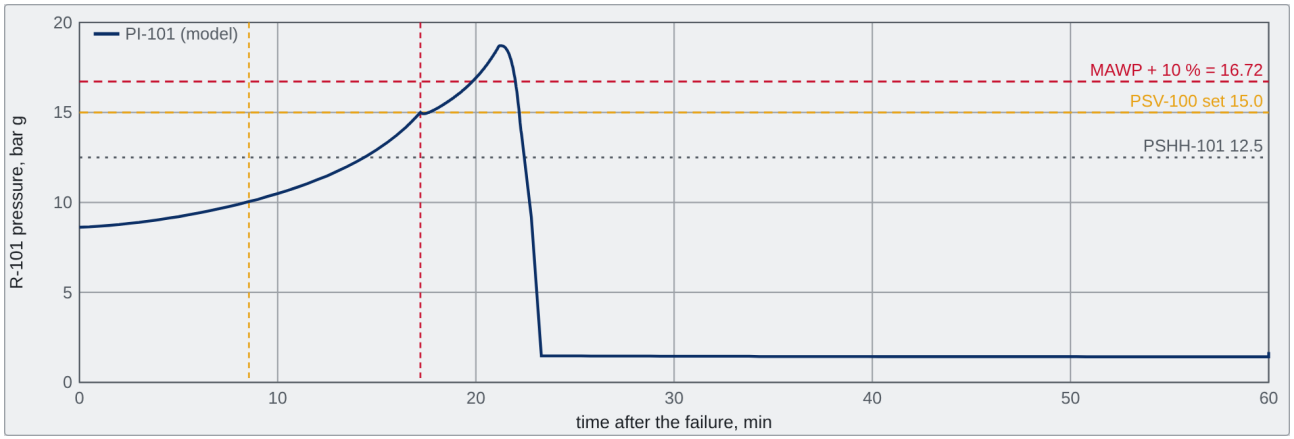


Figure 2 · Pressure in the governing run (Jacket temperature control fails to full steam (TV-110A wide open) at the design state), SIS off.

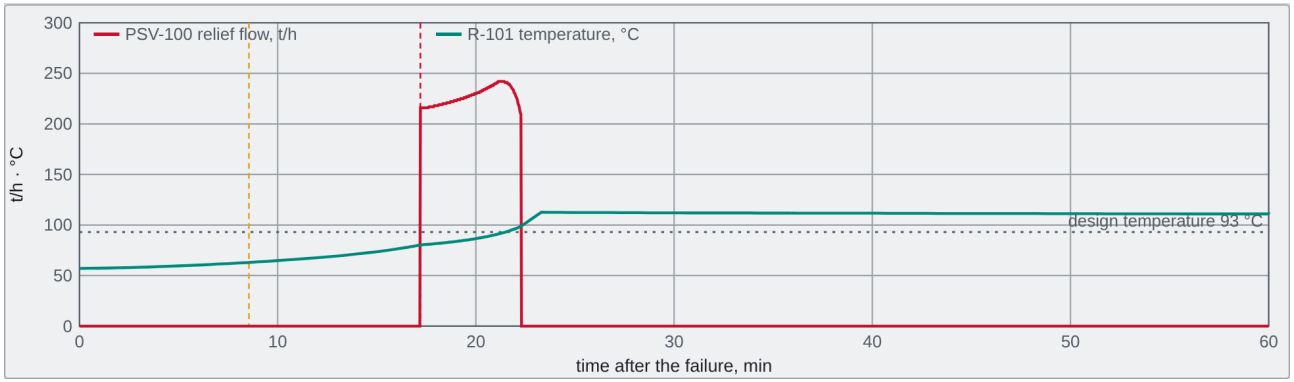


Figure 3 · Relief flow through the model PSV-100 and reactor temperature in the same run.

Compared with the SRS of SIF tab (SIF-1, same failure from the design state, trips off): TSHH-101 point 20:36 there, 20:35 here; PSHH-101 31:35 / 31:34; PSV-100 set 35:35 / 35:34 — the same model, so the process safety time (900 s) and the relief timing agree. After the lift the run goes on: peak 15.72 bar g, 14.7 t relieved in 243 s, then the pressure falls because the monomer is used up (polymerised and relieved).

5 Sizing (API 520 / 521)

5.1 PSV-100, vapour

Relief load W	345900 kg/h (TV-110A full open)
Relieving pressure P_1	16.50 bar g = 1751 kPa a (10 % overpressure)
Relieving temperature	85.4 °C (VCM + water vapour pressure = P_1)
M, k, Z	62.5; 1.18 (assumed, ideal gas); 0.82 (assumed, generalised chart)
Kd, Kb, Kc	0.975; 1; 0.9 (rupture disc RD-100 upstream)
C	0.02545
Required area	19182 mm²
Selected	2 × R (6 × 8 in, 20646 mm²), capacity 372 t/h
Model PSV-100 at P_1	227 t/h ≈ 12599 mm² (T)

5.2 External fire

Wetted area	56.9 m² (liquid height 8.00 m at the design state, bottom 4 m above grade assumed, 7.6 m limit → 3.60 m wetted)
Heat input	Q = 43.2·F·A^0.82 = 1188 kW (F = 1, adequate drainage and fire fighting)
Relieving	18.39 bar g (21 % accumulation), 90.5 °C, λ 228.5 kJ/kg
Load	18720 kg/h — PSV-100 passes 410 t/h at these conditions

5.3 Liquid and jacket

Liquid overfill (FV-103 wide open, 10.0 m³/h at the design pressure, G 0.996, Kd 0.65): A = 75 mm² → E. In the model the injection-pump shut-off stops the pressure at 14.98 bar g, below the set.

Blocked in (TSV-110). With both jacket valves closed and P-101 stopped the jacket water is heated by the batch: the model's no-circulation term gives Φ = $0.3 \cdot UA_j \cdot (T - T_{jo})$ = 414 kW at the design state. Thermal expansion (API 521): $q = \alpha \cdot \Phi / (\rho \cdot c_p)$ = 4.2e-4 × 414204 / (991 × 4180) = 151 L/h (150 kg/h; α typical for water at 45 °C). A liquid-full jacket gains α/κ ≈ 9.3 bar per kelvin (κ 4.5e-10 1/Pa, typical) while it heats at 0.59 K/min, so it reaches its 5.2 bar g design in ≈ 40 s from 1.5 bar g: a TSV is needed. Area 2.2 mm² → D (1 × 2 in), set 5.2 bar g, discharging to the CW return.

Blocked outlet. P-101 absorbs 90 kW (econ()) at 130 kg/s; with η 0.75 (assumed) the differential is 5.16 bar. With the CW return at 1.5 bar g (assumed) the jacket inlet already runs at ≈ 6.7 bar g, and with the return blocked the loop climbs to the CW supply (4 bar g, assumed) + shut-off head (1.2 × 5.16) ≈ 10.2 bar g — above the 5.2 bar g jacket design. A device for that case passes the full CW injection (TV-110B wide open, 110 kg/s ≈ 397 m³/h): 5829 mm² → Q, far more than a TSV. Recorded as an engineering query (part 8).

6 Inlet and outlet

Item	Input (assumed)	Result	Limit	✓/✗
Inlet line (one valve)	ID 202.7 mm, L 1 m, ΣK 0.5, f 0.014, 186148 kg/h	ΔP 16.3 kPa = 1.09 % of set	≤ 3 %	✓
Outlet line to D-103	ID 590.6 mm, L 30 m, ΣK 3, f 0.013, D-103 125 kPa a, 372296 kg/h	back-pressure 1.02 bar g, built-up 5.2 % of set	≤ 10 % conventional	✓

7 Depressuring and emergency venting

ESD (HS-001) from the design state trips the SIS and opens XV-108 to D-103: 33.6 t/h at 5 min, 5.63 bar g at 15 min (guideline ≤ 50 % of MAWP = 7.6 bar g), 3.45 bar g and 22.9 t vented after 59 min; an SIS trip without the vent gives 4.96 bar g (Transition operation tab, model runs).

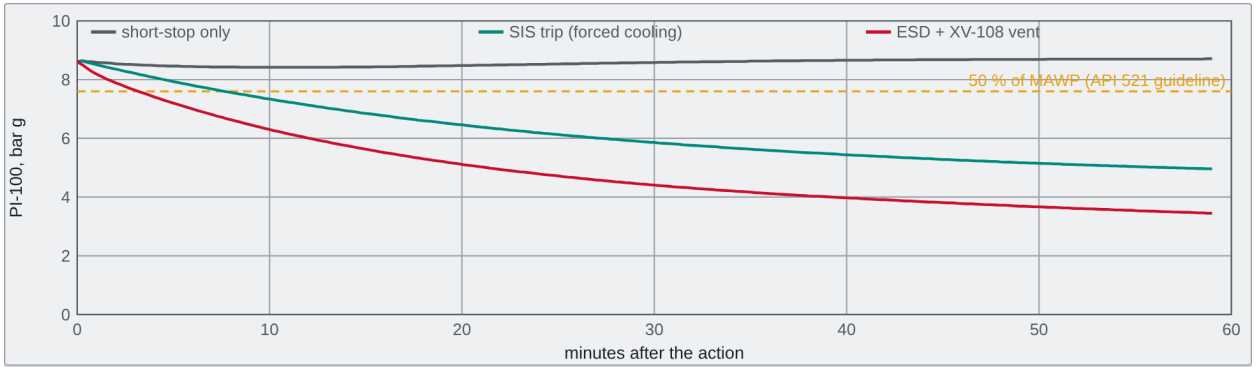


Figure 4 · PI-100 after a short-stop, an SIS trip and an ESD with the emergency vent (Transition operation tab runs).

• **D-103** receives PSV-100 and XV-108 (PFD-100 stream 20, C&E XV-108). It must hold one full relief without overfilling: vapour-only, the largest model relief is 26.3 t (CW loss · start, 29 m³ if condensed at 30 °C, ρ_{VCM} ≈ 893 kg/m³); the ESD vent sends 22.9 t in the first hour. If the relief is two-phase it can carry up to the whole liquid inventory, 121.3 m³ of slurry at the design state (117.8 t). Typical PVC practice is a dump / catch tank sized for one reactor charge with quench water to stop the reaction and a short-stop connection.

• **Vapour from D-103** must go to a closed destination — VCM recovery through the gasholder **D-106** (1.25 bar a, C-101 compressor) or a vent treatment — never to atmosphere (VCM is a carcinogen and extremely flammable; see the Electrical tab for its flammability data). PFD-100 does not show that route (query).

• **D-106** takes the degassing (HIC-105) and inert vents (HIC-135) in normal operation; a relief or ESD vent of 26 t VCM is a large slug for a gasholder at 1.25 bar a, so D-103 is the buffer that lets the recovery take it over hours.

• The relief line and D-103 must be free-draining towards D-103, inspected for polymer (HAZOP R35) and designed for the PSV-100 back-pressure of part 5.

8 Checks against the model and the other documents

17 ✓ · 4 ✗ (computed in the OTS page when this document was generated).

Check	Value	Criterion	✓/✗	Source
PSV-100 set ≤ R-101 MAWP	15.0 bar g	≤ 15.2 bar g (ASME VIII)	✓	K.Ppsv; model specification
Accumulation with the model PSV-100: CW loss · design	15.72 bar g	≤ 16.72 bar g (MAWP + 10 %)	✓	Model run, SIS off
Accumulation with the model PSV-100: CW loss · start	15.00 bar g	≤ 16.72 bar g (MAWP + 10 %)	✓	Model run, SIS off
Accumulation with the model PSV-100: TV-110A full open	18.71 bar g	≤ 16.72 bar g (MAWP + 10 %)	✗	Model run, SIS off
Accumulation with the model PSV-100: Power failure	15.00 bar g	≤ 16.72 bar g (MAWP + 10 %)	✓	Model run, SIS off
Fire: PSV-100 capacity at 21 % accumulation ≥ fire load	410 t/h vs 18.7 t/h	capacity ≥ load	✓	Calculators 4.1, 4.3
PSV-100 (2 × R) capacity ≥ worst model relief rate	372 vs 242 t/h	capacity ≥ rate	✓	Runs 3.x
PSV-100 (2 × R) capacity ≥ governing load at the set	372 vs 346 t/h	capacity ≥ load	✓	Run: TV-110A full open
Model PSV-100 (step(): 0.25·√(P − 1.2) kmol/s) ≥ governing load	227 vs 346 t/h	capacity ≥ load	✗	model.js step()
CW-loss run reproduces the SRS process-safety-time run: TSHH point	20:35 vs 20:36	within 30 s	✓	SRS tab SIF-1
… and PSV-100 set reached	35:34 vs 35:35	within 30 s	✓	SRS tab SIF-1
TSV needed on the blocked-in jacket	reaches 5.2 bar g in 40 s	TSV-110 in the register if < 10 min	✓	Calculation 4.4
Jacket design ≥ P-101 discharge in normal operation	6.7 bar g (derived)	≤ 5.2 bar g	✗	econ() 90 kW, η 0.75 and CW return 1.5 bar g assumed
PSV-100 / XV-108 discharge is closed (to D-103, not atmosphere)	R-101 → D-103 (PFD-100 stream 20, Safety layers)	closed destination	✓	PFD-100, SL_IT
D-103 vapour outlet shown on PFD-100	no stream leaves D-103	a route to recovery / treatment	✗	PFD-100
PSV-100 inlet loss (defaults of 5)	1.1 % of set	≤ 3 % (API 520 Part II)	✓	Calculator 5, assumed piping
PSV-100 built-up back-pressure (defaults of 5)	5.2 % of set	≤ 10 % conventional	✓	Calculator 5, assumed piping
ESD vent XV-108: pressure 15 min after the ESD	5.63 bar g	≤ 50 % of MAWP = 7.6 bar g (API 521 guideline, typical)	✓	Transition tab ESD run
Liquid overfill stays below the set (pump shut-off)	14.98 bar g	< 15.0 bar g	✓	Model run 3.7 (margin 0.02 bar)
EM_EQL R-101 design pressure ≥ PSV-100 set and = 15.2	15.2 bar g	≥ 15 and = 15.2	✓	Mechanical tab EM_EQL
EM_EQL D-103 design pressure ≥ PSV-100 back-pressure	7.0 vs 1.02 bar g	≥ back-pressure at the valve	✓	Mechanical tab EM_EQL, calculator 5

9 Engineering queries

#	Subject	Finding	Raised against	Proposed
ER-01	PSV-100 capacity in the model vs the relief load	The OTS model relieves PSV-100 with nPsv = 0.25·√(P − 1.2) kmol/s, i.e. 227 t/h at 10 % overpressure (≈ 12599 mm², one T orifice). The governing case (Jacket temperature control fails to full steam (TV-110A wide open) at the design state, SIS off) needs 346 t/h of vapour at the set, and in that run the model PSV lets R-101 reach 18.71 bar g, above MAWP + 10 % (16.72 bar g).	Model (step()) and PSV-100 sizing	Size PSV-100 for 346 t/h: 19182 mm² → 2 × R (6 × 8 in, conventional, with RD-100); update the relief coefficient in step() to the selected capacity (372 t/h at 16.5 bar g) and re-run the relief cases.
ER-02	Two-phase relief of the runaway (DIERS) not done	A VCM suspension relieves as a foamy two-phase slurry; the model and this basis use vapour only, which gives the smallest possible area. HAZOP R08 is still open.	PVC-V1-PE-RLB-001, HAZOP R08	Two-phase (DIERS, e.g. homogeneous-equilibrium / omega method) sizing of PSV-100 / emergency vent and of the line to D-103 from calorimetry data; keep the vapour figures as the lower bound.

#	Subject	Finding	Raised against	Proposed
ER-03	Jacket loop design pressure vs P-101	P-101 absorbs 90 kW (econ()) at 130 kg/s: with η 0.75 (assumed) that is a 5.2 bar differential, so the jacket inlet runs at ≈ 6.7 bar g (CW return 1.5 bar g assumed) and reaches ≈ 10.2 bar g with the outlet blocked (CW supply 4 bar g, shut-off 1.2 × head, assumed) — both above the 5.2 bar g jacket design (P&ID-001 note 3).	P&ID-001 note 3, Process hydraulics (P-101), Mechanical (jacket)	Confirm the P-101 head and the CW pressures; either raise the jacket design to cover CW supply + P-101 shut-off or add a jacket PSV for the blocked-outlet case (full CW injection 397 m³/h → 5829 mm², Q); TSV-110 (D) covers thermal expansion only.
ER-04	D-103 outlet and capacity	PFD-100 shows PSV-100 and XV-108 discharging to D-103 but no outlet from D-103. The model runs relieve up to 26.3 t (CW loss · start) as vapour and the ESD vent sends ≈ 23 t (Transition tab); a two-phase relief could carry up to the whole liquid inventory (121.3 m³ at the design state).	PFD-100, Mechanical (D-103), Relief basis	Show the D-103 vapour route (to VCM recovery / gasholder D-106 or vent treatment), size D-103 for one full two-phase relief plus quench water, and give D-103 its own design pressure ≥ the PSV-100 back-pressure.
ER-05	Temperature during a relieving runaway above the design temperature	With the SIS off the batch reaches 106.3 °C (CW loss, 4.85 bar g at that moment) and 112.6 °C (TV-110A full open, 1.48 bar g) — above the 93 °C design temperature of R-101.	Mechanical (R-101 design conditions)	Add a coincident design condition (pressure at the peak temperature) to the R-101 datasheet or confirm the vessel is acceptable at that temperature; this is an upset, not the design point.
ER-06	Short-stop pot D-105 pressure	D-105 must sit above the trip pressure (PSHH-101 12.5 bar g) plus line losses to inject; its pressure, its N_2 supply and any restriction orifice are not stated. If its N_2 supply exceeds the 15 bar g set, gas blow-through after the injection is a PSV-100 scenario.	P&ID-001, Process	State the D-105 operating / design pressure and the N_2 supply pressure; add a restriction orifice and include the blow-through case in the relief basis if the supply is above the set.
ER-07	Overfill: margin between the injection pump shut-off and the PSV set	With FV-103 wide open the vessel goes liquid-full and the pressure stops at the injection-pump shut-off (K.Pinj 16 bar a = 14.99 bar g): 14.98 bar g, only 0.02 bar below the set.	Process (injection pump), Relief basis	Confirm the injection-pump curve; if its shut-off can exceed 15 bar g, add the liquid case to PSV-100 (≈ 10 m³/h, see 4.2) and keep LSHH-100 independent (HAZOP R16) with a second shut-off valve (R17).
ER-08	Power failure reaches the PSV set after the SRS run window	The SRS run of power failure stops at 60 min (peak 10.39 bar g). The same case run for 3 h with the SIS off lifts PSV-100 at 77:32 (TSHH point 55:02, PSHH point 72:48).	SRS of SIF (PVC-V1-IN-SRS-001), SIF-2/SIF-3 basis	Extend the SRS demand run to the PSV set (≥ 90 min) and state the process safety time for power failure; essential power for M-101 and P-101A/B (HAZOP R02) stays the main measure.

10 Assumptions and references

• Assumed: k 1.18, Z 0.82 for VCM vapour; R-101 bottom 4 m above grade; CW supply 4 bar g, CW return 1.5 bar g; P-101 efficiency 0.75, shut-off 1.2 × rated head; N_2 header 8 bar g; water α 0.00042 1/K, κ 4.5e-10 1/Pa; inlet and outlet piping of section 6.

• Model: R-101 OTS model (model.js: K, step(), PsV, PsW, lamV, econ()), design state 2.0 h into the reaction; model relief runs of section 4 (tabs/erel/tools/compute.js).

• Other OTS documents: PFD-100, PVC-PID-001/002, Safety layers register, HAZOP report PVC-V1-SA-HZP-001 (R02, R08, R16, R17, R35, R39), SRS PVC-V1-IN-SRS-001 (SIF-1), Transition operation ESD runs.

• Standards quoted as typical practice: API 520 Part I (sizing) and Part II (installation), API 521 (scenarios, fire, depressuring), API 526 (orifice letters), ASME VIII Div. 1 (MAWP, accumulation), DIERS (two-phase relief).